# Syllabus

*MATH 1071, Section 001 – Intro to Discrete Mathematics* *Spring 2026*

**Instructor Name**: Ethan Davis
**Email**: davis5ej@mail.uc.edu
**Office Location**: French Hall West, Room 4513
**Office Hours (in the MASS Center – FH2133)**: Monday, 9AM – 12PM.

**Class Days & Times:** MWF 12:20PM – 1:15PM **Class Location:** Old Chem 665

***If you need to contact me**, email is my preferred method of communication. I do my best to respond to emails within 24 hours on a weekday and 48 hours on the weekend. Sometimes I miss an email or forget, so if more than 48 hours have passed, please send a follow-up!*

Students are expected to review and follow the linked course policies, university policies, and community resources on the following topics in the UC Student Resources Canvas site.

- Attendance
- Class Cancellation
- Academic Integrity
- Accessibility
- Title IX
- Counseling Services

## Course Overview

Math 1071 is a course designed for students interested in information technology and programming that includes topics in logic, number systems, set theory, methods of proof, probability, logic networks, and graph theory.

*Pre-requisites: MATH1022 min grade C-, or MATH1024 min grade C-, or MATH1026 min grade C-.*

**(Required) Textbook:** *Discrete Mathematics with Applications*, (5th edition) by Epp., with WebAssign.

The fee for the course materials (WebAssign access, including e-book) was included in your tuition. You can purchase a paper print version of the textbook from the UC bookstore at an additional cost (optional).

We will cover the following sections in the textbook

- **Chapter 1:** 1.2-1.3
- **Chapter 2:** 2.1-2.4
- **Chapter 3:** 3.2-3.3
- **Chapter 4:** 4.1-4.5,4.7-4.8
- **Chapter 5:** 5.1-5.3,5.6
- **Chapter 6:** 6.1-6.2
- **Chapter 7:** 7.1-7.2
- **Chapter 9:** 9.1-9.7
- **Chapter 10:** 10.1-10.2

## Learning Outcomes

Successful completion of the course means a student should be able to…

- Translate statements and arguments into symbolic form and construct truth tables to evaluate these statements and arguments.
- Apply basic logic to the analysis of digital logic circuits.
- Perform operations in various number systems and with 8-bit representation.
- Apply predicate logic to statements and arguments; apply rules of inference to analyze arguments; construct direct proofs, proofs by contradiction, and proofs by mathematical induction.
- Draw and apply Venn diagrams.
- Perform operations with matrices and vectors and use these in applications.
- Use combinations and permutations to find probabilities.
- State and apply the Binomial Theorem.
- Classify types of graphs, find paths, circuits, and construct minimum spanning trees.

## Course Format

This class is an **in-person** class. In-person class meetings will NOT be recorded. Students at the University of Cincinnati are expected to attend class to meet the course's learning objectives. **You are responsible for what we discuss in class, whether you attend or not.**

## Quizzes and Exams

There will be weekly 'quizzes' that will become available on Friday, and due Sunday. The lowest quiz grade will be dropped. **There are no make-up quizzes**, so a missed quiz will be counted as a 0 unless a valid reason is presented to the instructor prior to the quiz date or, if circumstances mandate, immediately afterward. Quizzes are given on canvas. You are welcome to work with other students on quizzes.

There will be three exams and a final exam. A **missed exam will count as a 0 unless a valid reason is presented to the instructor** prior to the exam date or, if circumstances mandate, immediately afterward.

| | |
|---|---|
| Exam 1 (Chapters 1,2,3) | Friday, February 6th |
| Exam 2 (Chapters 4,5) | Monday, March 9th |
| Exam 3 (Chapters 6,7,9) | Wednesday, April 15th |
| Final Exam (Cumulative) | Monday, April 27th (12:30PM – 2:30PM) |

You are **not** allowed to work with other students on exams. Using external resources like Chegg, Stack Exchange, ChatGPT, etc. on **ANY** assignments is considered academic misconduct.

## Homework

Homework sets will be electronically assigned and graded through Web Assign. There will be homework due on Sunday of each week. You are welcome to work with other students on homework. **Late Policy**: *you are allowed to make one 7-day extension request (per homework) on Web Assign and complete the homework late for 50% credit.*

## Grading

Your grade for this course is based on the following categories:

| | |
|---|---|
| Quizzes | 7.5% |
| Homework | 35% |
| Exams | 37.5% (3 exams, 12.5% each) |
| Final (Cumulative) | 20% |

**Drop Period Ends January 26 @ 11:59 pm EDT (*100% Refund and class removed from academic record)***

**Last day to Withdrawal Ends April 10 @ 11:59 pm EST (*No Refund and class remains on academic record as W)***

### Letter Grades

I expect to use the typical ten-point grading scale, with rounding at .5% as follows:

| Letter Grade | A | A- | B+ | B | B- | C+ | C | C- |
|---|---|---|---|---|---|---|---|---|
| Weighted Total Min | 92.5% | 89.5% | 86.5% | 82.5% | 79.5% | 76.5% | 72.5% | 69.5% |

There is no curving in this course, but borderline cases (within 1% from grade cutoff) *might* be considered for a rounding up.

**NP Grading Scheme**
Students who complete the course without earning at least a C- will be assigned the grade of NP (Not Proficient). An NP does not lower your GPA; however, it also does not count as a passing grade, earn course credit or meet any pre-requisites. If this course is an essential part of your academic plan and you earn an NP as your final grade, you must retake the course successfully in a future semester.

Students who receive an NP might have earned enough points for a D+, D, or D- grade. These students can request a change in their grade from NP for up to one year after the NP grade was assigned. It is important to consider that D grades will affect your GPA and might affect your academic standing. Contact your academic advisor to decide whether to request this change or if you have questions about the effect of the NP in your academic program.

**IMPORTANT:** Student must academically attend class to be eligible for an NP; this is especially important for students receiving financial aid. Attending just one or two class meetings does qualify as academic attendance.

More information about academic attendance can be found on Office of the Registrar Academic Attendance page.

## Academic Integrity

The University Rules, including the Student Code of Conduct and other documented policies of the department, college and university related to academic integrity including the assessment policies listed above will be enforced. https://www.uc.edu/campus-life/conduct/student-code-of-conduct.html

**The sanction for Academic Misconduct will be failure of the course.**

(AI) Artificial Intelligence/Apps/Extensions

You should not use artificial intelligence engines (e.g. ChatGPT), cell phone apps (e.g. Symbolab, Photomath), or browser extensions (e.g. Transcript, Coursology) to produce work for this class. While these might help with checking your solutions, if misused they often provide incorrect and incomplete information. You might trick yourself into believing your calculus ability is stronger than it really is or worse—confuse yourself by introducing advanced topics beyond the scope of this course. Besides, the majority of course credit comes from in-person assessments where these resources are strictly prohibited. Use of AI on the Quizzes, Homework, Exams, or Final Exam will be reported as Academic Misconduct.

**The sanction for Academic Misconduct will be failure of the course.**

If you are in doubt as to whether you are using an online learning support appropriately, I encourage you to meet with me for advice.

Please make note of and review the College of Arts and Science's Two-Strikes Policy. We take these violations seriously https://www.artsci.uc.edu/information/undergrad/academic-policies.html.

## Accessibility Policy

UC implemented a centralized platform to oversee the accommodation process called UC Accommodate. Students with special needs should register with Accessibility Resources and familiarize themselves with UC Accommodate platform as soon as possible to arrange reasonable provisions to ensure an equitable opportunity to meet all the requirements of this course.

The University of Cincinnati is committed to providing all students with equal access to learning opportunities. Accessibility Resources is the official campus office that works to arrange for reasonable accommodations for students with an identified physical, psychological or cognitive disability (learning, ADD/ADHD, psychological, visual, hearing, physical, cognitive, medical condition, etc.) Students are encouraged to contact the Accessibility Resources Office to arrange for a confidential meeting to discuss services and accommodations. Contact should be initiated as soon as possible to allow adequate time for accommodations to be arranged. Additional information can be found here: https://www.uc.edu/campus-life/accessibility-resources.html

## Student Religious Accommodations

Ohio law and the University's Student Religious Accommodations for Courses Policy 1.3.7 permits a student, upon request, to be absent for reasons of faith or religious or spiritual belief system or participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization and/or to receive alternative accommodations with regard to examinations and other course requirements due to an absence permitted for the above-described reasons. Not later than fourteen days after the first day of instruction in the course, a student should provide the instructor with written notice of the specific dates for which the student requests alternative accommodations. For additional information about this policy, please contact the Executive Director of the Office of Equal Opportunity and Access at (513) 556-5503 or oeohelp@UCMAIL.UC.EDU.

## Communication of Changes

The professor reserves the right to make changes to the syllabus, including assignment due dates and test dates (excluding any officially scheduled examinations) when necessary to meet learning objectives or when unforeseen circumstances occur. These changes will be announced as early as possible so students can adjust their schedules. Be assured that I will communicate to you any changes to our schedule, syllabus or policies quickly and efficiently through **Canvas.** It is your responsibility to make sure Canvas has a valid email address and it is recommended that you check this email daily. All course documents will be available via link in Canvas as well.

## Intellectual Property Statement

All original instructor-provided content for this course, which includes handouts, assignments, lectures, and video content is the intellectual property of the instructor(s). Students enrolled in the course this academic term may use the original instructor-provided content for their learning and completion of course requirements this term, but such content must not be distributed or monetized in any way.